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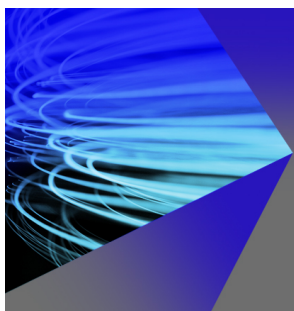
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ABSTRACT

Accurately predicting ionic conductivity is key to accelerating the discovery of solid-state electrolytes; yet, high-throughput experiments are labor-intensive, and first-principles simulations remain too slow. Previously, machine-learning models have been developed that incorporate structural, potential, or compositional information for prediction. Still, their poor prediction accuracy for the distinct ionic conductivities of polymorphic materials with identical compositions but different crystal structures remains problematic. Here, we present a three-dimensional convolutional neural network that learns from both the three-dimensional geometry and the spatial potential landscape of lithium ions. The former is expressed by the lithium-ion distribution, and the latter is encoded from the Bond Valence Sum Energy to predict ionic conductivity. The proposed model, trained on experimental ionic conductivities, achieves prediction performance comparable to that of a state-of-the-art model that uses composition information, and it ranks the mobilities of several materials sharing the same composition but different structures. Our findings highlight the effectiveness of incorporating structural information into conductivity prediction models to overcome the limitations of conventional approaches, potentially improving the screening of solid electrolytes by utilizing the three-dimensional physical descriptor.

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I. INTRODUCTION

The development of high-performance solid electrolytes is critical for realizing next-generation energy storage technologies, particularly all-solid-state lithium (Li)-ion batteries. These materials must exhibit high Li-ion conductivity (σ) to overcome the safety, leakage, and stability issues associated with conventional liquid electrolytes, while also supporting high energy density and long-term operation.¹ Experimental evaluation of σ , however, requires complex and resource-intensive procedures, including material synthesis, shaping, and electrochemical measurements.² Although *ab initio* molecular dynamics (AIMD) simulations can provide accurate estimates of conductivity, the high computational cost—often tens to hundreds of CPU hours per material—makes AIMD impractical for high-throughput materials discovery.³

To address these limitations, machine-learning (ML) approaches have been increasingly adopted to accelerate the prediction of material properties from structural and/or compositional information. For the prediction of σ , an ML model utilizing only chemical composition (Compositionally Restricted Attention-Based Network: CrabNet⁴) has achieved high regression accuracy.⁵ However, because it does not use structural information, it is impossible to distinguish between structurally distinct phases of the same compound. This becomes a critical issue, particularly in polymorphic and phase-transition-prone materials. In fact, not only composition but also structural characteristics play an important role in the physical properties of battery materials; many materials are known whose conductivities and capacities differ greatly across structural phases.^{6,7} As a result, CrabNet fails to capture essential structure-dependent conductivity behavior.

Structure-based ML methods employing graph neural network (GNN) architectures have been extensively developed.^{8–11} These pioneering models represent crystalline materials as graphs, where atoms are nodes and interatomic interactions are edges. A promising approach was taken for the prediction of σ using GNNs, in which the local minima of the potential energy of Li were represented by nodes and the diffusion paths between them were represented by edges.¹² However, a fundamental limitation of GNNs is that they do not guarantee invariance in crystal structures; namely, their outputs are likely to differ depending on the unit-cell choice. For example, the graph generated from a $1 \times 1 \times 1$ unit cell differs from that generated from a $2 \times 2 \times 2$ supercell, despite the fact that both cells represent an identical crystal because of periodic boundary conditions. This lack of unit-cell invariance is particularly problematic when long-range ion transport mechanisms define σ , which is commonly the case.

To overcome the limitations of both graph-based and composition-based models, we propose a structure-aware neural network framework that directly incorporates the energy landscape relevant to Li-ion transport. We use the Bond Valence Sum Energy (BVSE) method, which provides a three-dimensional potential energy distribution associated with Li-ion migration.¹³ This potential map, combined with the three-dimensional distribution of Li ions, is then converted into reciprocal lattice space and discretized into a voxel representation known as the Reciprocal 3D Voxel Space (R3DVS) descriptor.¹⁴ This descriptor is able to express long-range structures of the potential distribution due to the conversion into reciprocal lattice space, and it is invariant to unit-cell choice and translation of real-space distributions, thereby overcoming the limitations of prediction models using GNNs. The R3DVS descriptors of the BVSE and the distribution of Li ions are combined

as the input to a three-dimensional convolutional neural network (3DCNN). The model is able to learn spatially resolved, physically meaningful features related to low-energy diffusion pathways of Li and activation barriers from the three-dimensional structural data. Our model can distinguish between different structural phases of the same composition, unlike composition-based models such as CrabNet, enabling it to account for polymorphism and other phase-dependent conduction phenomena.

Similar approaches have been reported. For example, Bond Valence Sum (BVS) has been used to predict σ in oxygen-ion conductors.¹⁵ In addition, a combination of the Coulomb potential and Born–Mayer repulsion has been applied to predict σ in olivine-type and thiophosphate Li-ion conductors.¹⁶ Our approach is comparatively specific to the conduction pathways of ions by using BVSE and the Li distribution. We also utilize experimental values of σ when training our prediction model, unlike many previous studies.^{12,16} Although experimental values include errors and uncertainties of the experimental measurements, they incorporate every mechanism of ion diffusion, including long-range transport, ion hopping, and the effects of grain boundaries.

II. METHODS

The overview of the flow of the Li-ion conductivity (σ) prediction is shown in Fig. 1(a). The three-dimensional distributions of the BVSE and the concentrations of Li ions are converted into the R3DVS descriptor, and this is used as the set of inputs for the 3DCNN, which predicts σ . Two examples of the cross section of the R3DVS data are shown in Fig. 1(b). The two data sets are those of β -Li₃PS₄ (left) and γ -Li₃PS₄ (right), which are polymorphs with the same composition. As described by Kajita *et al.*,¹⁴ the grid loca-

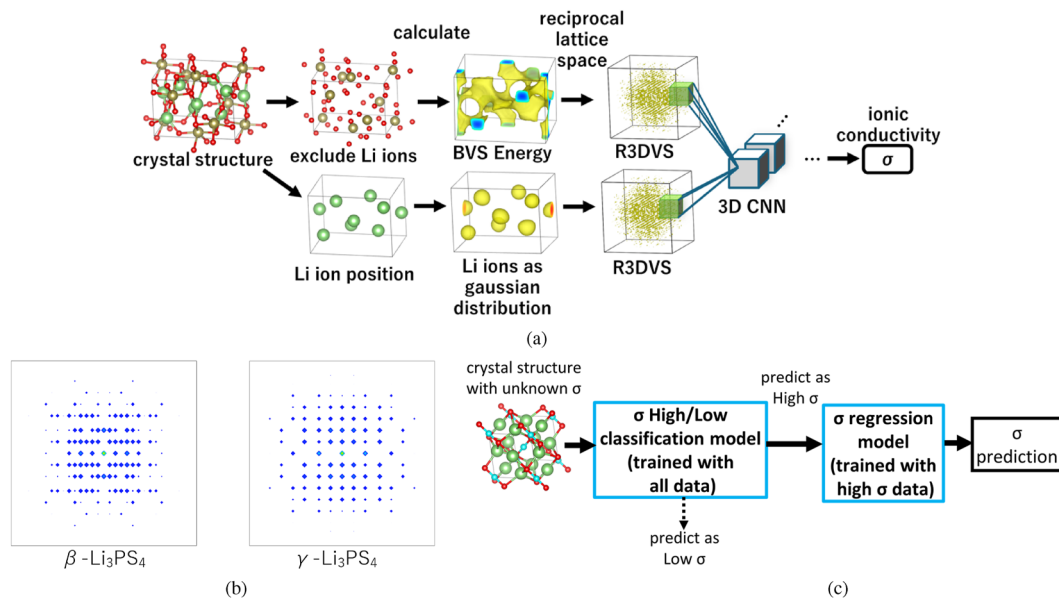


FIG. 1. (a) A schematic illustration of the flow of the prediction of ionic conductivity using 3DCNN. (b) The R3DVS of β and γ -Li₃PS₄ displayed on cross-sections passing through the origin of their reciprocal lattice spaces. (c) A schematic illustration of the two-step σ prediction model, comprising two 3DCNN models. Each model is trained according to the flow in (a). For data visualization, VESTA¹⁷ is used.

tions at which patterns arise in the R3DVS are determined by crystal structure symmetry.

Our dataset consists of 455 compounds with experimental $\sigma > 10^{-8}$ S/cm and 88 compounds with experimental $\sigma \leq 10^{-8}$ S/cm, resulting in a total of 543 compounds. Because the values of the lower σ records are not very reliable due to measurement errors, we propose a two-step process to predict σ [Fig. 1(c)]. We develop a classification model that determines whether the input crystal structure exhibits a σ higher or lower than 10^{-8} S/cm, and a regression model that predicts the value of σ for potential high Li-ion conductors.

In the following sections, the preparation of the descriptors and the procedure for training the 3DCNN are described. In addition, the training of the CrabNet model⁴—which takes the chemical compound of materials as input—with the same dataset used in this research is described for a fair comparison with the proposed 3DCNN model.

A. Preparation of the dataset

We compiled crystal structures and corresponding experimental Li-ion conductivities at room temperature from the Inorganic Crystal Structure Database (ICSD¹⁸) and relevant literature sources.^{6,19–21} For compounds for which ionic conductivities are known but structural data are unavailable, we generated hypothetical structures by substituting elements into known prototype structures. The list of prototype structures is provided in the [supplementary material](#). The lattice parameters of these hypothetical structures were scaled according to the ratio of the average ionic radii between the substituted and prototype elements. The final dataset consists of 543 compounds, including 248 generated structures.

B. Descriptors

The ionic conductivity σ can be described as²²

$$\sigma = \frac{z^2 F^2 c}{RT} D_0 \exp\left(-\frac{E_a}{RT}\right), \quad (1)$$

where z is the charge number of the ion (+1 for Li ions), F is the Faraday constant, c is the concentration of the ionic species, R is the universal gas constant, T is the absolute temperature, and E_a is the activation energy for diffusion. D_0 , the pre-exponential factor, depends on the vibration frequency of ions, which determines the occurrence rate of conduction, the dimension and number of conduction pathways, and other situational factors such as concerted hopping or site-occupancy disorders. To accurately predict the Li-ion conductivity, the descriptors must capture information related to E_a , D_0 , and c . In this study, the effects of E_a and D_0 are partially incorporated by introducing the energy landscape of Li ions. The spatial distribution of Li ions is introduced to account for c .

The energy landscape is computed using the BVSE method,¹³ which is a classical force field approach. This approach is widely recognized for its ability to approximate Li-ion migration barriers with relatively low computational cost.¹³ In BVSE, the attractive interaction between ions is modeled by a Morse-type potential, while repulsive interactions are treated as Coulombic. The Morse-type potential between two ions is given by

$$U_{\text{Morse}}(\mathbf{r}, \mathbf{r}_i) = D_e \left\{ \left[e^{\alpha(r_e - |\mathbf{r} - \mathbf{r}_i|)} - 1 \right]^2 - 1 \right\}, \quad (2)$$

where $\mathbf{r}$ and $\mathbf{r}_i$ are the positions of the ions, and r_e , D_e , and α are empirical parameters. The Coulomb potential between two ions is expressed as

$$U_{\text{Coulomb}}(\mathbf{r}, \mathbf{r}_i) = \frac{q_1 q_2}{|\mathbf{r} - \mathbf{r}_i|} \operatorname{erfc}\left(\frac{|\mathbf{r} - \mathbf{r}_i|}{\rho}\right), \quad (3)$$

where q_1 and q_2 denote the partial charges on the two ions. The complementary error function $\operatorname{erfc}(x)$ introduces a screening effect to account for the attenuation of the Coulomb interaction at short range. The factor ρ is the screening factor, which modulates the range of the interaction. All force field parameters are determined based on the paper by Adams and Rao.¹³ The resulting BVSE potential distribution of Li ions is, therefore,

$$U_{\text{BVSE}}(\mathbf{r}) = \sum_i [U_{\text{Morse}}(\mathbf{r}, \mathbf{r}_i) + U_{\text{Coulomb}}(\mathbf{r}, \mathbf{r}_i)], \quad (4)$$

where i is summed over all the ions in the unit cell, excluding Li ions. To construct the Li-ion energy landscape, every Li ion is initially removed from the crystal structure. A single Li ion is then systematically inserted into the structure at intervals of 0.2 Å throughout the lattice, and $U_{\text{BVSE}}(\mathbf{r})$ is computed for each position.

Because the Li ions are removed in the calculation, the resulting $U_{\text{BVSE}}(\mathbf{r})$ lacks explicit information about the original Li-ion concentration c , which is known to have a significant effect on ionic conductivity²³ as can be seen from Eq. (1). To recover the information regarding the Li-ion distribution, a Gaussian broadening is applied to the original Li-ion positions, thereby reconstructing a continuous Li-ion density profile. The Li density profile p at position $\mathbf{r}$ is described as

$$p(\mathbf{r}) = \sum_j \frac{\chi_j}{\sqrt{8\pi^3} d^3} \exp\left(-\frac{|\mathbf{r} - \mathbf{r}_j|^2}{2d^2}\right), \quad (5)$$

where χ_j is the occupancy of the j th Li ion, $\mathbf{r}_j$ is the position of the j th Li ion, and d is the broadening parameter of the Li ion. d is set to 0.6 Å. The index j is summed over the Li ions in the supercell of $3 \times 3 \times 3$, and the position $\mathbf{r}$ of the center unit cell is extracted. The obtained Li-ion density profile is also used as an explanatory variable for the prediction of σ .

The resulting three-dimensional distributions of both $U_{\text{BVSE}}(\mathbf{r})$ and $p(\mathbf{r})$ were converted into the R3DVS descriptor¹⁴ for input into the 3DCNN. By mapping real-space information into reciprocal lattice space, the R3DVS descriptor preserves translational symmetry and ensures invariance with respect to the choice of unit cell. However, rotational symmetry is not inherently preserved in this representation. To address this limitation, data augmentation was performed by generating 50 randomly rotated versions of each material's R3DVS descriptor. This strategy allows the 3DCNN to learn rotationally invariant features during training. The detailed procedure for confirming rotational invariance is provided in the [supplementary material](#).

The parameters used to generate the R3DVS descriptors are $\delta L = 0.4$ Å and $L = 25.6$ Å. The voxel size of the R3DVS descriptors is 64^3 . Details of the parameters are found in Ref. 14.

C. Training of the 3DCNN models

To predict the ionic conductivity, we propose a two-step modeling framework, as illustrated in Fig. 1(c). In the first step, a binary classification model determines whether the input structure exhibits σ greater than 10^{-8} S/cm. If the classification result is positive, the structure proceeds to the second step, where a regression model is used to predict the value of σ . If the classification result is negative, the structure is assumed to have $\sigma \leq 10^{-8}$ S/cm.

To ensure unbiased evaluation, 12 compounds with the same chemical formula but different structures (Li_3PS_4 , $\text{Li}_7\text{La}_3\text{Hf}_2\text{O}_{12}$, $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$, $\text{Li}_{0.45}\text{La}_{0.52}\text{TiO}_3$, $\text{Li}_{0.27}\text{La}_{0.58}\text{TiO}_3$, and Li_8GeP_4 , each with two distinct structural phases) are excluded from the training data in both models. These excluded materials, which have $\sigma > 10^{-8}$ S/cm, are reserved for testing the regression performance of the 3DCNN model. The binary classification model is trained using 531 compounds in the dataset. In contrast, the regression model is trained only on compounds with experimentally measured $\sigma > 10^{-8}$ S/cm, resulting in 443 compounds.

TABLE I. (a) 3DCNN structure of the binary classification model. (b) 3DCNN structure of the regression model. The slope α of the activation function Leaky ReLU was set to 0.3.

(a)	
Layer	Parameters
3D covolution	Kernel size: $6 \times 6 \times 6$ 32 filters
Leaky ReLU	$\alpha = 0.3$
Max pooling	Kernel size: $2 \times 2 \times 2$
3D covolution	Kernel size: $6 \times 6 \times 6$ 32 filters
Leaky ReLU	$\alpha = 0.3$
Max pooling	Kernel size: $2 \times 2 \times 2$
Global average Pooling	
Dense (128)	
Leaky ReLU	$\alpha = 0.3$
Dense (64)	
Leaky ReLU	$\alpha = 0.3$
Dense (2)	
Softmax	
(b)	
Layer	Parameters
3D covolution	Kernel size: $6 \times 6 \times 6$ 64 filters
Leaky ReLU	$\alpha = 0.3$
Max pooling	Kernel size: $2 \times 2 \times 2$
Global average pooling	
Dense (128)	
Leaky ReLU	$\alpha = 0.3$
Dense (64)	
Leaky ReLU	$\alpha = 0.3$
Dense (1)	
Linear	

The structure of the 3DCNN for the binary classification model is shown in Table I (a), and the structure of the regression model is shown in Table I (b). In both models, the R3DVS of the BVSE and the Li distribution are stacked as two channels for input into the 3DCNN. The regression model was trained using $\log \sigma$. The layer structure and hyperparameters of the 3DCNN are optimized to obtain the best prediction accuracy on the test dataset. As can be seen in Fig. 1(b), differences in crystal symmetry lead to different R3DVS patterns. The kernel size that exhibited the best accuracy for both classification and regression was $6 \times 6 \times 6$. This indicates that it is effective for the model to learn mid-range spatial correlations with the $6 \times 6 \times 6$ kernel rather than just short-range features with the $3 \times 3 \times 3$ kernel; however, the longer-range correlations that can be specifically learned with the $9 \times 9 \times 9$ kernel are unnecessary.

The classification model with two convolutional and pooling layers performed better than the model with one of each layer, while the regression model performed better with one rather than two of each layer. This indicates that the classification task may depend on combinations of wavevector features spanning a wide range. On the other hand, accurate prediction of the σ value in the regression model may require preservation of finer wavevector features, which would be partially lost in a deeper model. Similarly, the regression model with 64 filters achieved slightly better predictive performance than the model with 32 filters, suggesting that increasing the number of filters enables the network to learn a larger variety of features without reducing the spatial resolution of the input.

We applied 10-fold cross-validation for training the 3DCNN model. The dataset was divided into 10 subsets. In each fold, 8 subsets were used for training, 1 for validation, and 1 for testing. This process was repeated 10 times, each with a different combination of training, validation, and test sets, resulting in 10 distinct models. The performance of the models is assessed by the average prediction accuracy on the test data.

D. Training of CrabNet

For direct comparison with our 3DCNN model, we also conducted 10-fold cross-validation of the CrabNet model using the same subsets for training, validation, and test data. CrabNet is a transformer-architecture model that includes a self-attention mechanism and is used to learn chemical properties from composition, that is, the atomic elements and their proportions. The CrabNet model for training was obtained from Ref. 24.

III. RESULTS AND DISCUSSIONS

A. Prediction of Li ion conductivity

The average accuracy of the binary classification task of the 3DCNN was 86.6% in the 10-fold cross-validation, as shown in Table II. Note that classification accuracy, however, is often insufficient to evaluate the model performance correctly due to the imbalanced data distribution. In fact, the present dataset consists of 443 compounds with $\sigma > 10^{-8}$ S/cm and 88 compounds with $\sigma \leq 10^{-8}$ S/cm. If a classification model predicts all materials to have $\sigma > 10^{-8}$, the accuracy will be 83%, although its predictions are not meaningful. The Matthews correlation coefficient (MCC) is an alternative metric that can assess the performance of binary classification

TABLE II. The prediction performance of classification and regression using 3DCNN and CrabNet. The values shown are the average predictions on the test data in the 10-fold cross-validation.

	Classification accuracy (%)	Classification MCC	Regression MAE	Regression R ²
3DCNN	86.6	0.496	0.810	0.386
CrabNet	89.0	0.549	0.748	0.462
Ensemble	...	...	0.711	0.520

even with imbalanced datasets, unlike accuracy or the F1 score.^{25,26} The MCC is defined as

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}}$$

TP : True Positive, TN : True Negative, FP : False Positive, FN : False Negative.

MCC = 1 shows a completely correct prediction, and MCC = -1 shows a completely incorrect prediction. When every prediction is positive or negative ($TN = FN = 0$ or $TP = FP = 0$), MCC = 0, meaning that the model is no better than a random prediction. We achieved 0.496 for the average MCC value of the 10-fold cross-validation of the 3DCNN, which indicates sufficient accuracy for a meaningful classification for the two-step prediction.

Next, we assess the 3DCNN regression model. Figure 2(a) shows the predicted $\log \sigma$ of the test data in the 10-fold cross-validation with respect to the experimental value. The average value of the mean absolute error (MAE) of the prediction of the test data is 0.810, and the R² value is 0.386. The results suggest that the model effectively learns the characteristics of the BVSE distribution and Li ion distribution to predict σ .

The prediction results of the trained CrabNet models are shown in Fig. 2(b). The prediction performances are summarized in

Table II. While the binary classification performance of the CrabNet model is similar to the results of the 3DCNN, the regression performance of CrabNet exceeds that of the 3DCNN. This indicates that the chemical information extracted from the composition is important in the prediction of σ as well, in addition to the structural information.

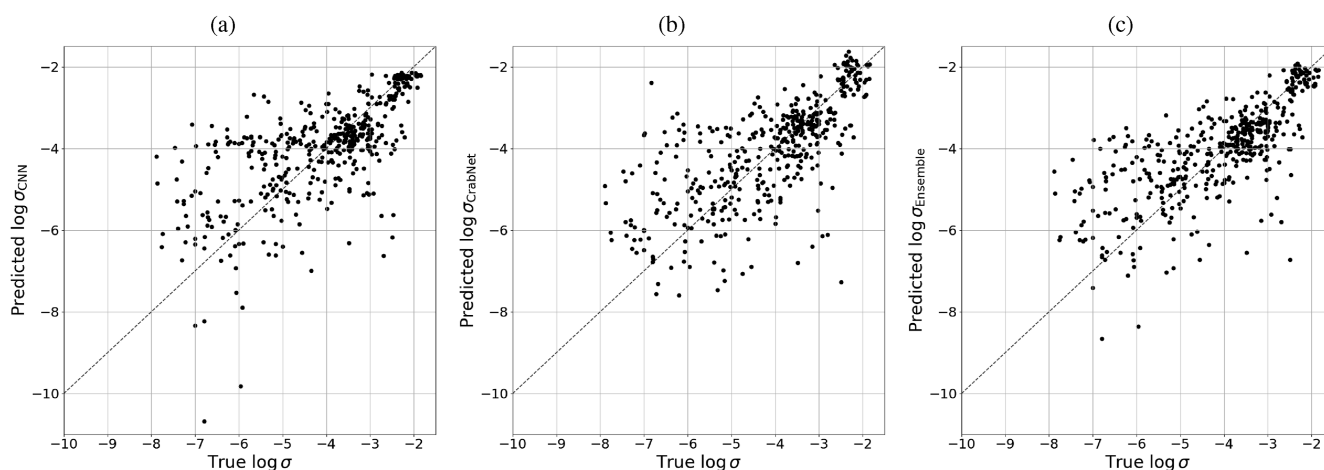
From the obtained results, the chemical composition and three-dimensional BVSE potential, which are distinct types of information, are both effective in the prediction of σ . When several prediction models with distinct predictors show similar prediction accuracies, ensembles typically show better performance.^{15,27} We adopt this approach and assess the prediction performance of the average value of the $\log \sigma$ predicted by the 3DCNN and CrabNet. We define $\log \sigma_{\text{Ensemble}}$ as

$$\log \sigma_{\text{Ensemble}} = \frac{\log \sigma_{\text{CNN}} + \log \sigma_{\text{CrabNet}}}{2}. \quad (7)$$

The predicted $\log \sigma_{\text{Ensemble}}$ is plotted in Fig. 2(c). We obtain a higher R² value and lower MAE than both the 3DCNN and CrabNet, as seen in Table II.

B. Prediction dependence on structure phases

As previously noted, the advantage of the present approach is its ability to distinguish different structural phases of the same composition. Here, we assess the structural dependence of σ using polymorphic materials. The regression results are shown in Fig. 3. The CrabNet model cannot distinguish between different structural phases, resulting in identical predictions. The 3DCNN outputs distinct predictions of polymorphs because it can recognize differences in crystal structures. As can be seen from the p-values of the polymorphs of Li₃PS₄, Li₇La₃Hf₂O₁₂, Li₇La₃Zr₂O₁₂, and Li_{0.45}La_{0.52}TiO₃, which are well below the common p-value threshold of 0.05, the 3DCNN model is able to accurately rank the polymorphs with larger/smaller ionic conductivities. On the other hand, the differences in the predicted σ of the polymorphs of

**FIG. 2.** Plot of the $\log \sigma$ values predicted by (a) the 3DCNN model, (b) the CrabNet model, and (c) the average value of the two predictions (ensemble) on the y-axis vs the experimental $\log \sigma$ values on the x-axis.

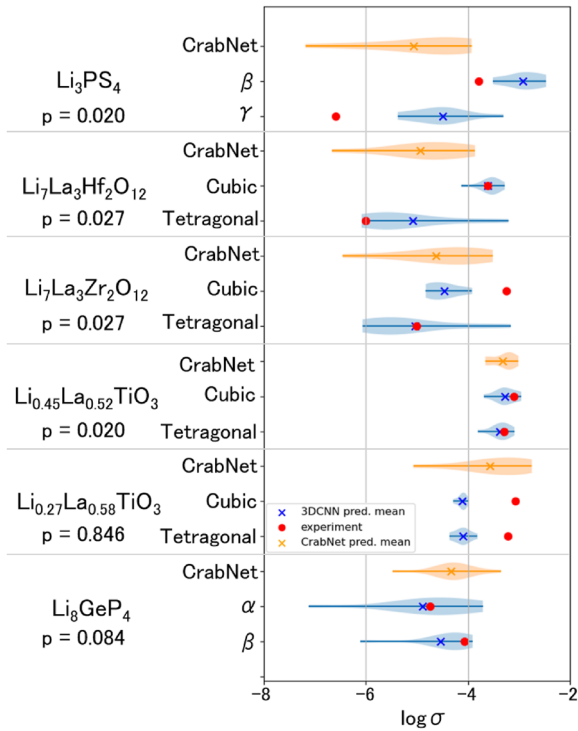


FIG. 3. Plots of the $\log \sigma$ values predicted by the 3DCNN model and the CrabNet model. The red dots indicate the experimental values of the ionic conductivity. The blue and orange violin plots represent the distributions of the predictions from the 10-fold cross-validation of the 3DCNN and CrabNet, respectively, and their average values are plotted with the blue and orange crosses. The p-values obtained from the Wilcoxon signed-rank test on the 10 paired predictions of structural phases are listed under their compositions.

$\text{Li}_{0.27}\text{La}_{0.58}\text{TiO}_3$ and Li_8GeP_4 are not statistically significant with respect to the p-value.

Next, we assess the advantage of the 3DCNN, which incorporates three-dimensional structural information, over the BVSE-calculated activation energy alone in evaluating changes in σ among polymorphs. The activation energy E_a is the barrier energy that Li ions need to overcome to migrate to a neighboring unit cell, and this contributes to the ionic conductivity exponentially, as expressed in Eq. (1). We approximate E_a using a calculation method based on the BVSE and denote this value as $\tilde{E}_a$; the detailed derivation is provided in the [supplementary material](#). The $\tilde{E}_a$ of the compounds are listed in [Table III](#).

Let us consider the ratio of σ between polymorphs (phase A and phase B) of the same composition. This can be expressed from Eq. (1) using the calculated $\tilde{E}_a$ as

$$\frac{\sigma_{\text{phase A}}}{\sigma_{\text{phase B}}} = \frac{c_{\text{phase A}} D_{0,\text{phase A}}}{c_{\text{phase B}} D_{0,\text{phase B}}} \exp\left(-\frac{\tilde{E}_{a,\text{phase A}} - \tilde{E}_{a,\text{phase B}}}{RT}\right). \quad (8)$$

If we naively assume that the effects of the pre-exponential factor D_0 and the Li ion concentration c do not change between phase transitions, the ratio of σ can be expressed as $\exp\left(-\frac{(\tilde{E}_{a,\text{phase A}} - \tilde{E}_{a,\text{phase B}})}{RT}\right)$. We compare this naive expression $\exp\left(-\frac{(\tilde{E}_{a,\text{phase A}} - \tilde{E}_{a,\text{phase B}})}{RT}\right)$, the ratio of the 3DCNN prediction $\sigma_{\text{phase A}}^{\text{pred.}}/\sigma_{\text{phase B}}^{\text{pred.}}$, and the ratio of the experimental value $\sigma_{\text{phase A}}^{\text{exp.}}/\sigma_{\text{phase B}}^{\text{exp.}}$ listed in [Table III](#). The ratios of the predicted σ are closer, in order of magnitude, to those of the experimental σ than the values expected from the $\tilde{E}_a$ differences alone. This suggests that the 3DCNN captures part of the pre-factor-related contributions, such as the frequency of Li migration and the Li ion concentration, in addition to the Arrhenius factor of the activation energy, as expressed in Eq. (1).

TABLE III. The predictions of different structural phases of polymorphic materials. The $\tilde{E}_a$ values calculated from the BVSE, the ratio of the predicted σ of polymorphs, and the ratio of the experimental σ of polymorphs are also listed. The notation "phase A" refers to the structural phase listed in the first row of each composition, and "phase B" refers to that of the second row.

Compound	$\tilde{E}_a$ (eV)	$\exp\left(-\frac{(\tilde{E}_{a,\text{phase A}} - \tilde{E}_{a,\text{phase B}})}{RT}\right)$	$\sigma_{\text{phase A}}^{\text{pred.}}/\sigma_{\text{phase B}}^{\text{pred.}}$	$\sigma_{\text{phase A}}^{\text{exp.}}/\sigma_{\text{phase B}}^{\text{exp.}}$
$\beta - \text{Li}_3\text{PS}_4$	0.490	2.71×10^4	37.5	6.15×10^2
$\gamma - \text{Li}_3\text{PS}_4$	0.755			
Cubic $\text{Li}_7\text{La}_3\text{Hf}_2\text{O}_{12}$	0.831	6.56	30.1	2.44×10^2
Tetragonal $\text{Li}_7\text{La}_3\text{Hf}_2\text{O}_{12}$	0.879			
Cubic $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$	0.942	1.37	3.77	57.5
Tetragonal $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$	0.950			
Cubic $\text{Li}_{0.45}\text{La}_{0.52}\text{TiO}_3$	1.279	1.14×10^2	1.28	1.57
Tetragonal $\text{Li}_{0.45}\text{La}_{0.52}\text{TiO}_3$	1.401			
Cubic $\text{Li}_{0.27}\text{La}_{0.58}\text{TiO}_3$	0.936	3.38×10^7	0.963	1.39
Tetragonal $\text{Li}_{0.27}\text{La}_{0.58}\text{TiO}_3$	1.385			
$\alpha - \text{Li}_8\text{GeP}_4$	0.770	33.5	0.446	0.209
$\beta - \text{Li}_8\text{GeP}_4$	0.861			

The remaining discrepancies between the predicted and experimental conductivity ratios may partly arise from physical effects that are not fully represented in the descriptors of the 3DCNN. For example, when the Li ion sites are fully occupied, Li ion migration must occur by pushing out the ion at its destination through Coulomb repulsion, which increases the effective barriers. On the other hand, partially occupied Li sites enable Li ions to migrate easily to a vacant site. These effects are especially prominent in the different phases of Li_8GeP_4 ,²⁸ $\text{Li}_7\text{La}_3\text{Hf}_2\text{O}_{12}$, and $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$,²⁹ thus, the precise inclusion of Li–Li interactions and partial occupancies may contribute to improved prediction accuracies of the 3DCNN.

IV. CONCLUSION

The Li-ion conductivities σ of solid materials were predicted using their three-dimensional Li-ion distributions and potential distributions. The 3DCNN model achieved prediction performance comparable to that of CrabNet, which predicts σ from only the chemical composition. The 3DCNN model was able to learn the structural features of the BVSE and Li-ion distributions related to factors governing σ . As a result, the 3DCNN model was able to distinguish and, in several cases, correctly rank polymorphic materials sharing the same composition.

There are multiple ways to potentially improve the prediction accuracy of σ . Including the effects of Coulomb repulsion between Li ions enables the expression of the potential landscape of Li ions to be more precise. In addition, in the current data, the occupancy factor of each atom is multiplied by the calculated terms of the BVSE or Li distribution, which only incompletely expresses the effects of partial occupancy in disordered structures on ionic conductivity. Therefore, for a more precise expression of partial occupancies, it is necessary to employ larger supercells. In addition, transfer learning using pre-trained 3DCNN models on larger datasets may further enhance performance, as implemented by Hargreaves *et al.*⁵ However, these improvements must be balanced against increased computational cost.

Overall, the study demonstrated the utility of structure-based descriptors, which provide information independent of chemical composition. These descriptors effectively enrich the representational space for predicting the ionic conductivity of solid electrolytes, laying a foundation for future improvements in screening materials for high- σ solid electrolytes through multi-scope descriptors and hybrid modeling strategies.

SUPPLEMENTARY MATERIAL

The [supplementary material](#) provides the procedure of structure generation, the confirmation of rotational symmetry of the 3DCNN, and the calculation of activation energy $\bar{E}_a$.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

M.H. implemented the training of the 3DCNN and CrabNet and prediction of materials. K.N. and H.Y. prepared the dataset and conducted the BVSE calculations. S.K., N.O., and H.Y. designed the research. All authors contributed to the analysis of results and writing of the manuscript.

Minori Hashizume: Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Koki Nakano:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Resources (equal); Software (equal); Writing – original draft (equal); Writing – review & editing (equal). **Nobuko Ohba:** Conceptualization (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Seiji Kajita:** Conceptualization (lead); Formal analysis (equal); Supervision (equal); Writing – review & editing (equal). **Hisatsugu Yamasaki:** Conceptualization (lead); Data curation (equal); Investigation (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data and codes used in this study are openly available in <https://github.com/ToyotaCRDL/ion-conductivity-3DCNN>.

REFERENCES

1. B. Goodenough and Y. Kim, “Challenges for rechargeable Li batteries,” *Chem. Mater.* **22**, 587–603 (2010).
2. W. D. Richards, L. J. Miara, Y. Wang, J. C. Kim, and G. Ceder, “Interface stability in solid-state batteries,” *Chem. Mater.* **28**, 266–273 (2016).
3. R. Sasaki, B. Gao, T. Hitosugi *et al.*, “Nonequilibrium molecular dynamics for accelerated computation of ion–ion correlated conductivity beyond Nernst–Einstein limitation,” *Npj Comput. Mater.* **9**, 48 (2023).
4. A. Y.-T. Wang, S. K. Kauwe, R. J. Murdock, and T. D. Sparks, “Compositionally restricted attention-based network for materials property predictions,” *npj Comput. Mater.* **7**, 77 (2021).
5. C. J. Hargreaves, M. W. Gaultois, L. M. Daniels, E. J. Watts, V. A. Kurlin, M. Moran, Y. Dang, R. Morris, A. Morscher, K. Thompson, M. A. Wright, B.-E. Prasad, F. Blanc, C. M. Collins, C. A. Crawford, B. B. Duff, J. Evans, J. Gamon, G. Han, B. T. Leube, H. Niu, A. J. Perez, A. Robinson, O. Rogan, P. M. Sharp, E. Shoko, M. Sonni, W. J. Thomas, A. Vasylenko, L. Wang, M. J. Rosseinsky, and M. S. Dyer, “A database of experimentally measured lithium solid electrolyte conductivities evaluated with machine learning,” *npj Comput. Mater.* **9**, 9 (2023).
6. K. Homma, M. Yonemura, T. Kobayashi, M. Nagao, M. Hirayama, and R. Kanno, “Crystal structure and phase transitions of the lithium ionic conductor Li_3PS_4 ,” *Solid State Ionics* **182**, 53–58 (2011).
7. T.-C. Liu, A. Salgado-Casanova, S. Yubuchi, B. Baldassarri, M. Aykol, J. Yoshida, H. Yamasaki, Y. Zhu, S. B. Torrisi, and C. Wolverton, “Tailored ordering enables high-capacity cathode materials,” [arXiv:2506.12545](#) (2025).
8. T. Xie and J. C. Grossman, “Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties,” *Phys. Rev. Lett.* **120**, 145301 (2018).
9. K. T. Schütt, P.-J. Kindermans, H. E. Sauceda, S. Chmiela, A. Tkatchenko, and K.-R. Müller, “SchNet – A deep learning architecture for molecules and materials,” *J. Chem. Phys.* **148**, 241722 (2018).

- ¹⁰J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Neural message passing for quantum chemistry," in *Proceedings of the 34th International Conference on Machine Learning* (PMLR, 2017), Vol. 70, pp. 1263–1272.
- ¹¹C. Chen, W. Ye, Y. Zuo, C. Zheng, and S. P. Ong, "Graph networks as a universal machine learning framework for molecules and crystals," *Chem. Mater.* **31**, 3564–3572 (2019).
- ¹²S. R. Xie, S. J. Honrao, and J. W. Lawson, "High-throughput screening of Li solid-state electrolytes with bond valence methods and machine learning," *Chem. Mater.* **36**(19), 9320–9329 (2024).
- ¹³S. Adams and R. P. Rao, "High power lithium ion battery materials by computational design," *Phys. Status Solidi A* **208**, 1746–1753 (2011).
- ¹⁴S. Kajita, N. Ohba, R. Jinnouchi, and R. Asahi, "A universal 3D voxel descriptor for solid-state material informatics with deep convolutional neural networks," *Sci. Rep.* **7**, 16991 (2017).
- ¹⁵S. Kajita, N. Ohba, A. Suzumura, S. Tajima, and R. Asahi, "Discovery of superionic conductors by ensemble-scope descriptor," *NPG Asia Mater.* **12**, 31 (2020).
- ¹⁶A. France-Lanord, R. Asahi, B. Leblanc, J. Lee, and E. Wimmer, "Highly efficient evaluation of diffusion networks in Li ionic conductors using a 3D-corrugation descriptor," *Sci. Rep.* **9**, 15123 (2019).
- ¹⁷K. Momma and F. Izumi, "VESTA: A three-dimensional visualization system for electronic and structural analysis," *J. Appl. Crystallogr.* **41**, 653–658 (2008).
- ¹⁸F. I. Z. Karlsruhe, Inorganic crystal structure database (ICSD), <https://icsd.fiz-karlsruhe.de>, 2025, release 2025.1.
- ¹⁹F. A. L. Laskowski, D. B. McHaffie, and K. A. See, "Identification of potential solid-state Li-ion conductors with semi-supervised learning," *Energy Environ. Sci.* **16**, 1264–1276 (2023).
- ²⁰R. Murugan, V. Thangadurai, and W. Weppner, "Fast lithium ion conduction in garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$," *Angew. Chem., Int. Ed.* **46**, 7778–7781 (2007).
- ²¹J. Wolfenstine, E. Rangasamy, J. L. Allen, and J. Sakamoto, "High conductivity of dense tetragonal $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$," *J. Power Sources* **208**, 193–196 (2012).
- ²²A. K. Verma, A. S. Thorat, and J. K. Shah, "Estimating ionic conductivity of ionic liquids: Nernst–Einstein and Einstein formalisms," *J. Ion. Liq.* **4**, 100089 (2024).
- ²³R. Jalem, T. Aoyama, M. Nakayama, and M. Nogami, "Multivariate method-assisted ab initio study of olivine-type LiMXO_4 (main group M^{2+} – X^{3+} and M^{3+} – X^{4+}) compositions as potential solid electrolytes," *Chem. Mater.* **24**, 1357–1364 (2012).
- ²⁴A. Y.-T. Wang, Compositionally-restricted attention-based network (CrabNet)-GitHub, <https://github.com/anthony-wang/CrabNet>.
- ²⁵B. W. Matthews, "Comparison of the predicted and observed secondary structure of T4 phage lysozyme," *Biochim. Biophys. Acta, Protein Struct.* **405**, 442–451 (1975).
- ²⁶D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics* **21**, 6 (2020).
- ²⁷K. P. Murphy, *Machine Learning: A Probabilistic Perspective* (The MIT Press, 2012).
- ²⁸H. Eickhoff, S. Strangmüller, W. Klein, H. Kirchhain, C. Dietrich, W. G. Zeier, L. van Wüllen, and T. F. Fässler, "Lithium phosphidogermanates α - and β - Li_8GeP_4 —A novel compound class with mixed Li^+ ionic and electronic conductivity," *Chem. Mater.* **30**, 6440–6448 (2018).
- ²⁹X. He, Y. Zhu, and Y. Mo, "Origin of fast ion diffusion in super-ionic conductors," *Nat. Commun.* **8**, 15893 (2017).